Unit IV

Unit: 4

Database Security: Authentication, Authorization and access control, DAC, MAC and RBAC models, Intrusion detection, SQL injection.

Advanced topics: Object oriented and object relational databases, Logical databases, Web databases, Distributed databases, Data warehousing and data mining.

Unit - IV

Authentication Database authentication is the process of confirming that a user who is attempting to log into a database is authorized to do so and is only accorded the rights to perform activities that he or she has been authorized to do.

e.g Mobile authentication : asks a PIN, pattern, password etc

Computer authentication: username, password etc

Data Authorization :- It is a security mechanism used to determine user privileges or access levels related to system resources including computer programs, files, services, data etc.

During authorization, a system verifies an authenticated user's access rules and either grants or refuses resource access.

Access Control It is a security technique that regulates who or what can view or use resources in a computing environment.
There are two types of access control
* Physical
* Logical

Physical Access Control limits access to campuses, buildings, rooms etc.
Logical Access Control limits connections to computer networks, system files and data.

Access Control System perform identification authentication and authorization of users
e.g login - username, password, biometric scan etc

DAC (Discretionary Access Control)
This method is based on granting & Revoking privileges. Let us consider RDBMS.
There are two levels for assigning privileges
* Account level - At this level, the DBA specifies the particular privileges that each account holds independently of the relations in the database.

* The relation (or table) level – At this level, we can control the privilege to access each individual relation or view in the database.

Account level

Examples :- CREATE SCHEMA or
CREATE TABLE privilege
CREATE VIEW privilege
ALTER
DROP
MODIFY
SELECT
privilege

Relation level The granting & revoking of privileges follows an authorization model for discretionary previleges known as Access Matrix Model where
rows of Matrix M represents (users, accounts, programs)
& columns " " " (relations, records, columns, views, operations)
Each position M(i, j) in the matrix represents the types of privileges (read, write, update) that subject i holds on object j.

MAC (Mandatory Access Control)

DAC technique is for RDBMS.
This is an all or nothing method.
i.e. a user either has or does not have certain privilege.
An additional security policy MAC is used to classify data & users based on security classes. (for multilevel security)

Security classes

Top secret	(TS)	Subject & Object
Secret	(S)	
Confidential	(C)	
Unclassified	(U)	

Commonly used Model for Multilevel Security is Bell-LaPadula model which classify each subject & object into one of security classes.
Two restrictions are enforced on data access based on subject/object classification.
(i) A subject S is not allowed read access to an object O unless class (S) class (O). This is known as the simple security property.

(ii) A subject S is not allowed to write an object O

(i) Simple Security Property :- A U-user can not read S-data.

(ii) Star Property :- A S-user can not write U-data.

(iii) Strong Star Property :- A S-user can not write U-data and a U-user can not write S-data.

To incorporate multilevel security notions into the relational database model, it is common to consider attribute values and tuples as data objects. Hence, each attribute A is associated with a classification attribute C in the schema and each attribute value in a tuple is associated with a corresponding security classification.

RBAC (Role-Based Access Control)

It is an access control method that creates permission to end users depends on their role within the organization.
RBAC is generally used in conjunction with the principle of least privilege where the roles represented will only involve the least level of access needed to done the necessary job service or requirements.
RBAC methodology is based on a group of three primary rules that govern access to secured systems.
Role Assignment: Each transaction or operation can only be carried out if the user has assigned the suitable role. Roles can be assigned by an independent party or selected by user attempting to implement the action.
Role Authorization: The objective of role authorization is to provide that users can only consider a role for which they have been given the suitable authorization. When a user consider a role, they should do so with authorization from an administrator.

Transaction Authorization An operation can only be done if the user trying to complete the transaction possesses the suitable role.

Intrusion Detection System (IDS)

IDS observes network traffic for malicious transactions and sends immediate alerts when it is observed. It is a S/W that checks a network or system for malicious activities. The intrusion detector learning task is to build a predictive model capable of distinguishing between 'bad connections' and 'good connections'.

How does an IDS work?

* An IDS monitors the traffic on a computer network to detect any suspicious activity.
* It analyzes the data flowing through the network to look for patterns & signs of abnormal behaviour.
* The IDS compares the network activity to a set of predefined rules & patterns to identify any activity that might indicate an intrusion.
* If the IDS detects something that matches one of these patterns, it sends an alert to the system administrator.

* The system administrator can then investigate the alert and take action to prevent any damage or further intrusion.

Classification of IDS

* Network Intrusion Detection System (NIDS)
* Host Intrusion Detection System (HIDS)
* Protocol-based Intrusion Detection System (PIDS)
* Application Protocol-based Intrusion Detection System (APIDS)
* Hybrid Intrusion Detection System (HIDS)

Benefits of IDS

* Detects malicious activity
* Improves network performance
* Compliance requirements
* Provides insights

SQL Injection

It is a technique used to exploit user data through web page inputs by injecting SQL commands as statements. Basically, these statements can be used to manipulate the application's web server by malicious users.

* SQL injection is a code injection technique that might destroy your database.
* SQL injection is one of the most common web hacking techniques.
* SQL injection is the placement of malicious code in SQL statements via webpage Input.

Let us consider an example

```
txtUserId = getRequestString ("UserId");
txtSQL = "SELECT * FROM users where
          UserId = " + txtUserId;
```

If there is nothing to prevent a user from entering "wrong" input, the user can enter

UserId : [105 or 1=1]

Then, the SQL statement is

SELECT * FROM Users where UserId=105 or 1=1;

This statement is valid and will return all rows from the Users table since 1=1 is always TRUE.

* A hacker might get access to all the user names and passwords in a database by simply inserting 105 or 1=1 into the input field.

OBJECT ORIENTED DATABASE	OBJECT RELATIONAL DATABASE
A database that represents information in the form of objects as used in Object Oriented Programming	A database that depends on the Relational model and Object Oriented Database Model
Depends on OOP	Depends on relational model and object oriented database model
Less improved	More improved

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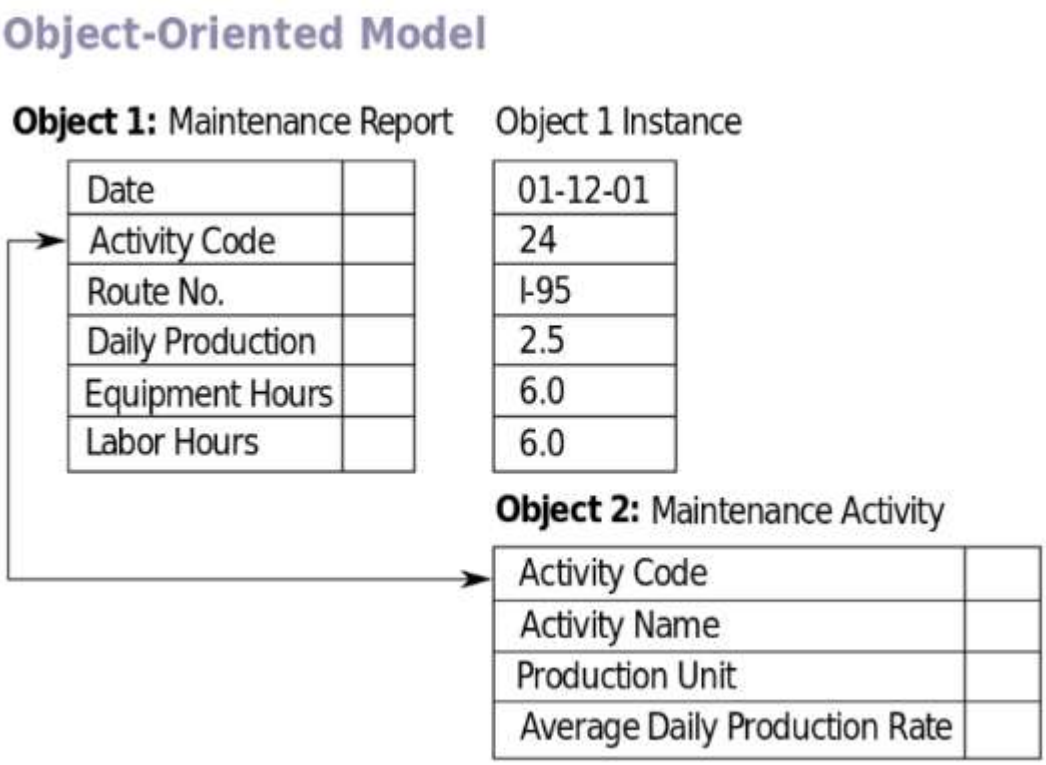
What is Object Oriented Database

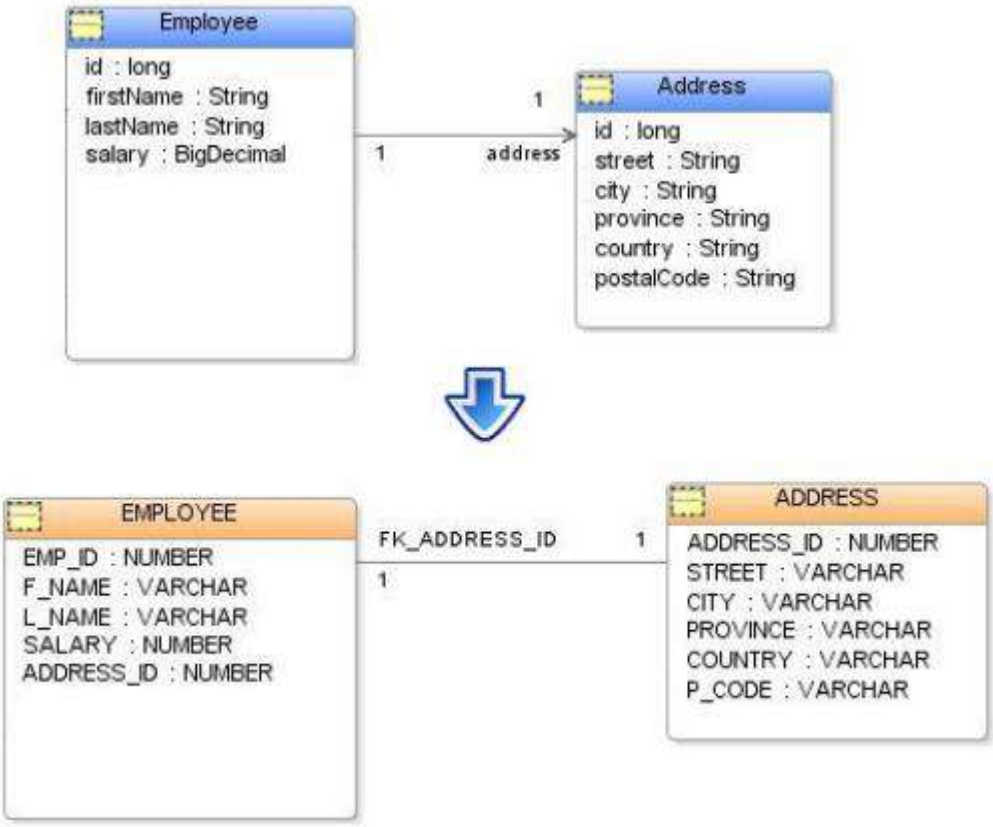
Object-oriented databases represent data in the form of objects and classes. According to the object-oriented paradigm, an object is a real-world entity. In addition, a class helps to create objects. Moreover, object-oriented databases follow the principles of object-oriented programming.



In addition, object-oriented databases support OOP concepts such as inheritance, encapsulation, etc. It also supports complex objects such as maps, sets, lists, tuples or collections of multiple primitive objects. Furthermore, Object-oriented database allows the user to create persistent objects which help to overcome the database issues such as concurrency and recovery. These objects stay in computer memory even after completing the execution.

What is Object Relational Database

Object-relational database is an improved version of the object-oriented database. It provides a solution to the issues users face in object-oriented databases; some of these issues include cost for computing resources, possibilities of design errors and data inconsistency. However, it is also important to note that object-relational database is based on both the relational model and the object-oriented database model.

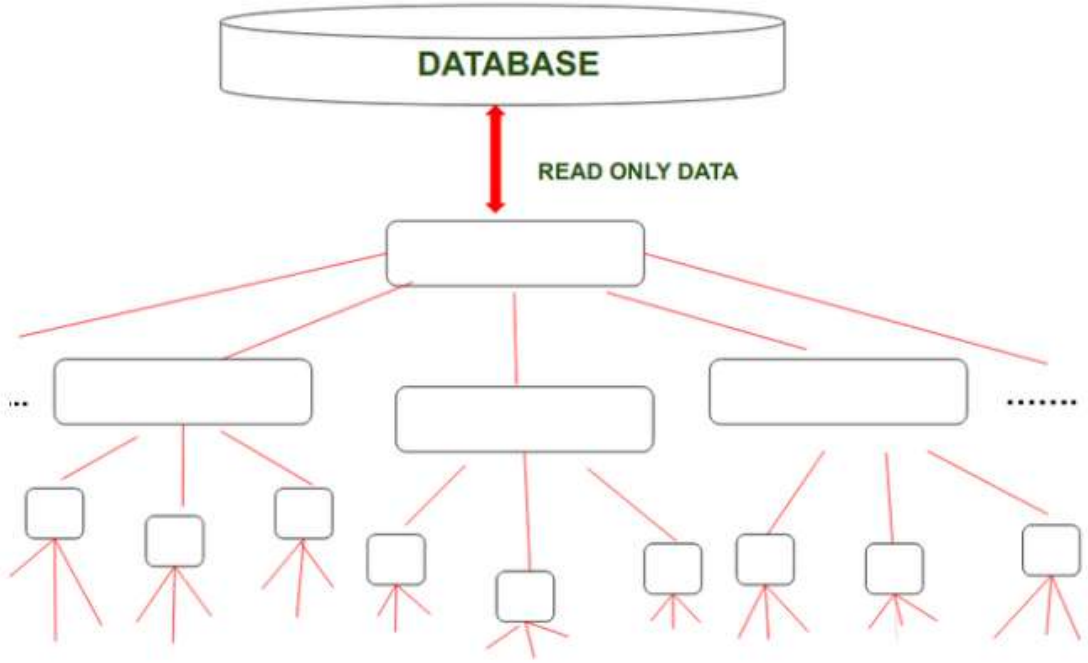


Furthermore, these databases support objects and inheritance and provide a better interface for many object-oriented languages. Users can also use data model extensions with custom data types and methods. Moreover, companies such as Microsoft, Oracle, and Sybase have object-relational versions of their products. These databases are more applicable for applications that consist of a large number of short-lived transactions on data items with complicated structures.

Criteria	Relational Database	Object Oriented Database
Definition	Data is stored in tables which consist of rows and columns.	Data is stored in objects. Objects contain data.
Amount of data	It can handle large amounts of data.	It can handle larger and complex data.
Type of data	Relational database has single type of data.	It can handle different types of data.
How data is stored	Data is stored in the form of tables (having rows and columns).	Data is stored in the form of objects.
Data Manipulation Language	DML is as powerful as relational algebra. Such as SQL, QUEL and QBE.	DML is incorporated into object-oriented programming languages, such as C++, C#.
Learning	Learning relational database is a bit complex.	Object oriented databases are easier to learn as compared to relational database.
Structure	It does not provide a persistent storage structure because all relations are implemented as separate files.	It provides persistent storage for objects (having complex structure) as it uses indexing technique to find the pages that store the object.
Constraints	Relational model has key constraints, domain constraints, referential integrity and entity integrity constraints.	To check the integrity constraints is a basic problem in object-oriented database.
Cost	The maintenance cost of relational database may be lower than the cost of expertise required development and integration of object oriented database.	In some cases hardware and software cost of object oriented databases is lower cost than relational databases.

Logical databases

A Logical Database is used to retrieve data from various tables and the data is interrelated to each other. Also, a logical database provides a read-only view of Data. A Logical database uses only a hierarchical structure of tables i.e. Data is organized in a Tree-like Structure and the data is stored as records that are connected to each other through edges (Links).The logical database reads the program, stores them in the program if required, and passes them line by line to the application program.



Tasks Of Logical Database:

Below is some important task of Logical Database:

- With the help of the Logical database, we will read the same data from multiple programs.
- A logical database defines the same user interface for multiple programs.
- Logical Database ensures the Authorization checks for the centralized sensitive database.
- With the help of a Logical Database, Performance is improved. Like in Logical Database we will use joins instead of multiple SELECT statements, which will improve response time and this will increase the Performance of Logical Database.

Data View Of Logical Database:

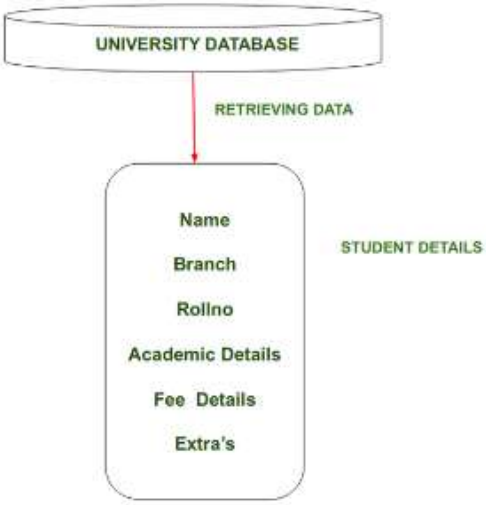
Logical Database provides a particular view of Logical Database tables. A logical database is appropriately used when the structure of the Database is Large. It is convenient to use flow i.e

- SELECT
- READ
- PROCESS
- DISPLAY

In order to work with databases efficiently. The data of the Logical Database is hierarchical in nature. The tables are linked to each other in a Foreign Key relationship.

Example:

Suppose in a University or College, a HOD wants to get information about a specific student. So for that, he firstly retrieves the data about its batch and Branch from a large amount of Data, and he will easily get information about the required Student but didn't alter the information about it.



Advantages Of Logical Database:

Let us look at some advantages of the logical database:

- In a Logical database, we can select meaningful data from a large amount of data.
- Logical Database consists of Central Authorization which checks for Database Accesses is Authenticated or not.
- In this Coding, the part is less required to retrieve data from the database as compared to Other Databases.
- Access performance of reading data from the hierarchical structure of the Database is good.
- Easy to understand user interfaces.
- Logical Database firstly check functions which further check that user input is complete, correct, and plausible.

Disadvantages Of Logical Database:

This section shows the disadvantages of the logical database:

- Logical Database takes more time when the required data is at the last because if that table which is required at the lowest level then firstly all upper-level tables should be read which takes more time and this slows down the performance.
- In Logical Database ENDGET command doesn't exist due to this the code block associated with an event ends with the next event statement.

Web Databases

The Web-based database management system is one of the essential parts of DBMS and is used to store web application data. A web-based Database management system is used to handle those databases that are having data regarding E-commerce, E-business, blogs, e-mail, and other online applications.

Requirements for Web DBMS Integration

Here are the lists of some of the most significant necessities for the database integration applications within the Web. These requirements are standards and not fully attainable at present. There is no ranking of orders, and so the requirements are as follows:

- The ability and right to use valuable corporate data in a fully secured manner.
- Provides data and vendor's autonomous connectivity that allows freedom of choice in selecting the DBMS for present and future use.
- The capability to interface to the database, independent of any proprietary Web browser and/or Web server.
- A connectivity solution that takes benefit of all the features of an organization's DBMS.
- An open-architectural structure that allows interoperability with a variety of systems and technologies; such as:
    - Different types of Web servers
    - Microsoft's Distributed Common Object Model (DCOM) / Common Object Model (COM)
    - CORBA / IIOP
    - Java / RMI which is Remote Method Invocation
    - XML (Extensible Markup Language)
    - Various Web services (SOAP, UDDI, etc.)
- A cost-reducing way which allows for scalability, development, and changes in strategic directions and helps lessen the costs of developing and maintaining those applications
- Provides support for transactions that span multiple HTTP requests.
- Gives minimal administration overhead.

Benefits of the Web-DBMS Approach

Here are various benefits that come through the use of web-based DBMS are:

- Provides simplicity
- Web-DBMS is Platform independence
- Provides Graphical User Interface (GUI)
- Standardization
- Provides Cross-platform support
- Facilitates transparent network access
- Scalability
- Innovation

Distributed Databases

A distributed database is basically a database that is not limited to one system, it is spread over different sites, i.e, on multiple computers or over a network of computers. A distributed database system is located on various sites that don't share physical components. This may be required when a particular database needs to be accessed by various users globally. It needs to be managed such that for the users it looks like one single database.

Types of Distributed databases:

1. Homogeneous Database:

In a homogeneous database, all different sites store database identically. The operating system, database management system, and the data structures used – all are the same at all sites. Hence, they're easy to manage.

2. Heterogeneous Database:

In a heterogeneous distributed database, different sites can use different schema and software that can lead to problems in query processing and transactions. Also, a particular site might be completely unaware of the other sites. Different computers may use a different operating system, different database application. They may even use different data models for the database. Hence, translations are required for different sites to communicate.

Applications of Distributed Database:

- It is used in Corporate Management Information System.
- It is used in multimedia applications.
- Used in Military's control system, Hotel chains etc.
- It is also used in manufacturing control system.

There are several different architectures for distributed database systems, including:

Client-server architecture: In this architecture, clients connect to a central server, which manages the distributed database system. The server is responsible for coordinating transactions, managing data storage, and providing access control.

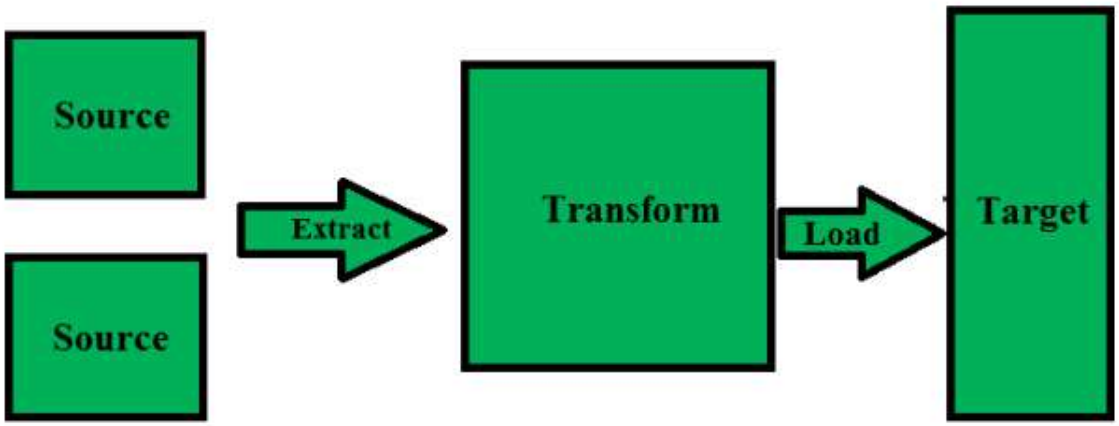
Peer-to-peer architecture: In this architecture, each site in the distributed database system is connected to all other sites. Each site is responsible for managing its own data and coordinating transactions with other sites.

Federated architecture: In this architecture, each site in the distributed database system maintains its own independent database, but the databases are integrated through a middleware layer that provides a common interface for accessing and querying the data.

Distributed database systems can be used in a variety of applications, including e-commerce, financial services, and telecommunications. However, designing and managing a distributed database system can be complex and requires careful consideration of factors such as data distribution, replication, and consistency.

Data Warehousing:

It is a technology that aggregates structured data from one or more sources so that it can be compared and analyzed rather than transaction processing. A **data warehouse** is designed to support the management decision-making process by providing a platform for data cleaning, data integration, and data consolidation. A data warehouse contains subject-oriented, integrated, time-variant, and non-volatile data. The Data warehouse consolidates data from many sources while ensuring data quality, consistency, and accuracy. It improves system performance by separating analytics processing from transnational databases. Data flows into a data warehouse from the various databases. A data warehouse works by organizing data into a schema that describes the layout and type of data. Query tools analyze the data tables using schema.



Advantages of Data Warehousing:

- The data warehouse's job is to make any form of corporate data easier to understand. The majority of the user's job will consist of inputting raw data.
- The capacity to update continuously and frequently is the key benefit of this technology. As a result, data warehouses are perfect for organizations and entrepreneurs who want to stay current with their target audience and customers.
- It makes data more accessible to businesses and organizations.
- A data warehouse holds a large volume of historical data that users can use to evaluate different periods and trends in order to create predictions for the future.

Disadvantages of Data Warehousing:

- There is a great risk of accumulating irrelevant and useless data. Data loss and erasure are other potential issues.
- Data is gathered from various sources in a data warehouse. Cleansing and transformation of the data are required. This could be a difficult task.

Data Mining:

It is the process of finding patterns and correlations within large data sets to identify relationships between data. Data mining tools allow a business organization to predict customer behavior. Data Mining tools are used to build risk models and detect fraud. Data mining is used in market analysis and management, fraud detection, corporate analysis, and risk management.

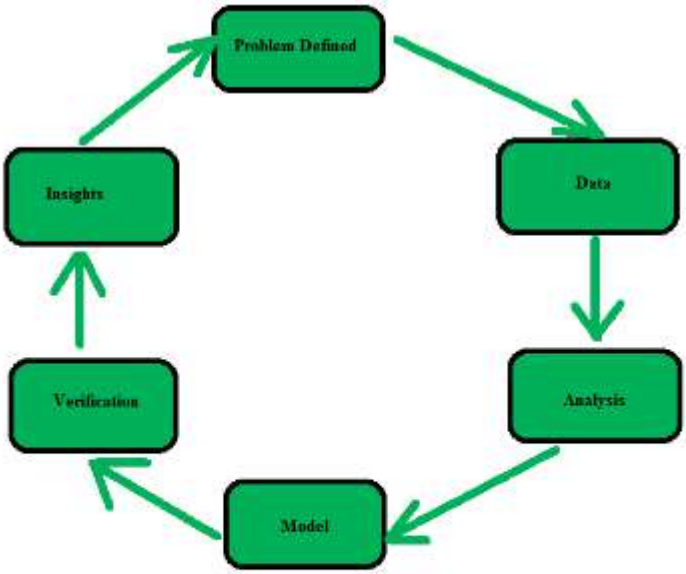


Figure: Data Mining process

Advantages of Data Mining:

- Data mining aids in a variety of data analysis and sorting procedures. The identification and detection of any undesired fault in a system is one of the best implementations here. This method permits any dangers to be eliminated sooner.
- In comparison to other statistical data applications, data mining methods are both cost-effective and efficient.
- Companies can take advantage of this analytical tool by providing appropriate and easily accessible knowledge-based data.
- The detection and identification of undesirable faults that occur in the system are one of the most astonishing data mining techniques.

Disadvantages of Data Mining:

- Data mining isn't always 100 percent accurate, and if done incorrectly, it can lead to data breaches.
- Organizations must devote a significant amount of resources to training and implementation. Furthermore, the algorithms used in the creation of data mining tools cause them to work in different ways.